19 Lecture 19 (February 28, 2025, est.): The Frobenius functor

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Throughout this lecture, k is algebraically closed of characteristic $p > 0$, and $\mathcal{C}$ is a semisimple symmetric tensor category. We write $\overline{\mathcal{A}}$ for the semisimplification of a symmetric category $\mathcal{A}$ by negligible morphisms. For the cyclic group C_p, let J_r be the indecomposable representation of dimension r, and let L_r be its image in Ver_p. Thus $L_p = 0$ and $L_1 = \mathbf{1}$.

19.1 Equivariant objects and semisimplification

If a finite group G acts trivially on $\mathcal{C}$, an object of $\mathcal{C}^G$ is a pair (X, ρ) with $\rho : G \to \mathrm{Aut}_{\mathcal{C}}(X)$. The morphisms commute with the specified actions. There is a symmetric equivalence

$$\mathcal{C}^G \simeq \mathcal{C} \boxtimes \mathrm{Rep}_k(G).$$

Indeed, write $X = \bigoplus_S S \otimes_k M_S$, where S runs through the simple objects of $\mathcal{C}$ and $M_S = \mathrm{Hom}(S, X)$. Since $\mathrm{End}(S) = k$ and there are no morphisms between distinct simples, the G-action is precisely a representation on each multiplicity space M_S.

19.1 Proposition. *There is a symmetric equivalence*

$$\overline{\mathcal{C}^G} \simeq \mathcal{C} \boxtimes \overline{\mathrm{Rep}_k(G)}.$$

Proof. The indecomposables of $\mathcal{C}^G$ are $S \boxtimes M$, with S simple and M an *indecomposable* G-module. Their categorical dimensions are $\dim(S)\dim_k(M)$. A simple object in a semisimple rigid category has nonzero categorical dimension: the two one-dimensional spaces $\mathrm{Hom}(\mathbf{1}, S \otimes S^*)$ and $\mathrm{Hom}(S \otimes S^*, \mathbf{1})$ pair nondegenerately, because the unit occurs with multiplicity one. Consequently $S \boxtimes M$ has dimension zero exactly when M does. The trace pairing on the S-isotypic block is the ordinary trace pairing on the multiplicity spaces multiplied by the nonzero scalar $\dim(S)$. Thus taking its radical, and then the quotient, acts separately on each G-module block. This gives the claimed equivalence, compatibly with tensor products and symmetry. ∎

19.2 Remark. It is simple objects, rather than all nonzero objects, that have nonzero dimension here. For example, $\mathbf{1}^{\oplus p}$ is nonzero but has dimension zero in characteristic p.

19.2 Tensor powers and additivity

The symmetry defines an action of S_p on $X^{\otimes p}$. Restrict it to $C_p = \langle \sigma \rangle$, where $\sigma = (1\,2\,\cdots\,p)$. This gives a functor

$$P : \mathcal{C} \longrightarrow \mathcal{C}^{C_p}, \qquad X \longmapsto (X^{\otimes p}, \sigma), \quad f \longmapsto f^{\otimes p}.$$

The canonical regrouping of tensor factors makes P symmetric monoidal, but P is not additive. Compose it with semisimplification to obtain

$$\mathrm{Fr}_0 : \mathcal{C} \longrightarrow \overline{\mathcal{C}^{C_p}} \simeq \mathcal{C} \boxtimes \mathsf{Ver}_p.$$

19.3 Lemma. *The functor* Fr_0 *is additive.*

Proof. For $f, g : X \to Y$, expand $(f+g)^{\otimes p}$. Besides $f^{\otimes p}$ and $g^{\otimes p}$, every term is a word in f and g containing both letters. The cyclic group acts freely on these words, since p is prime. The sum over one orbit has the form

$$T = \sum_{i=0}^{p-1} \sigma_Y^i h \sigma_X^{-i}, \qquad h = h_1 \otimes \cdots \otimes h_p.$$

It is C_p-equivariant. For an equivariant map $u : Y^{\otimes p} \to X^{\otimes p}$, naturality and cyclicity of the trace give

$$\mathrm{Tr}(\sigma_Y^i h \sigma_X^{-i} u) = \mathrm{Tr}(h \sigma_X^{-i} u \sigma_Y^i) = \mathrm{Tr}(hu).$$

Hence $\mathrm{Tr}(Tu) = p\,\mathrm{Tr}(hu) = 0$. Each orbit sum is negligible, so the mixed terms vanish after semisimplification. This proves additivity on morphisms. Since $P(0) = 0$, preservation of biproducts follows as well. ∎

19.3 The scalar twist

Although additive, Fr_0 satisfies $\mathrm{Fr}_0(\lambda f) = \lambda^p \mathrm{Fr}_0(f)$. Define $\mathcal{C}^{(1)}$ to have the same underlying additive symmetric monoidal category as $\mathcal{C}$, with scalar multiplication

$$\lambda \cdot_{(1)} f = \lambda^p f.$$

Because k is perfect, this is again a tensor category over k. The category Ver_p is defined over $\mathbb{F}_p$, and therefore $\mathsf{Ver}_p^{(1)} \simeq \mathsf{Ver}_p$.

19.4 Definition. The *Frobenius functor* is the resulting k-linear symmetric monoidal functor

$$\mathrm{Fr} : \mathcal{C} \longrightarrow \mathcal{C}^{(1)} \boxtimes \mathsf{Ver}_p.$$

It is exact because every exact sequence in $\mathcal{C}$ splits. This argument uses semisimplicity; exactness will be a genuine condition in the last lecture. The construction and additivity argument are due to Ostrik [Ost15, Section 3.3].

19.4 The example Ver_5^+

Let $X = L_3$ in Ver_5^+. Its fusion rule is $X^{\otimes 2} = \mathbf{1} \oplus X$, so

$$X^{\otimes 5} = \mathbf{1}^{\oplus 3} \oplus X^{\oplus 5}.$$

As an equivariant object this has the form $\mathbf{1} \boxtimes M_1 \oplus X \boxtimes M_X$, with $\dim_k M_1 = 3$ and $\dim_k M_X = 5$. Put $\varphi = (1+\sqrt{5})/2 = \mathrm{FPdim}(X)$. Tensor functors between fusion categories preserve Frobenius–Perron dimensions, hence

$$\varphi = \mathrm{FPdim}(\overline{M_1}) + \varphi\,\mathrm{FPdim}(\overline{M_X}).$$

The possible decompositions of a three-dimensional C_5-module are J_3, $J_2 \oplus J_1$, and $J_1^{\oplus 3}$. Their semisimplifications have Frobenius–Perron dimensions φ, $\varphi + 1$, and 3, respectively. Thus equality forces $M_1 = J_3$ and $\overline{M_X} = 0$. Since M_X has dimension five, it must be J_5. Therefore

$$\mathrm{Fr}(X) = \mathbf{1} \boxtimes L_3.$$

19.5 The induced operation on the Grothendieck ring

Write $K_0(\mathcal{C})$ for the Grothendieck ring; a Frobenius twist does not change this ring. Categorical dimension defines a ring map

$$\phi : K_0(\mathcal{C}) \otimes_{\mathbb{Z}} K_0(\mathsf{Ver}_p) \longrightarrow K_0(\mathcal{C}) \otimes_{\mathbb{Z}} k, \qquad [Y] \otimes [L_r] \longmapsto [Y] \otimes r.$$

19.5 Lemma. *For every $X \in \mathcal{C}$, one has $\phi([\mathrm{Fr}(X)]) = [X]^p$.*

Proof. Decompose $P(X)$ as $\bigoplus_S S \boxtimes M_S$. Forgetting the cyclic action gives $[X]^p = \sum_S \dim_k(M_S)[S]$. On semisimplifying, each J_r with $r < p$ contributes categorical dimension r, while J_p is killed and its dimension is already zero in k. The two expressions agree. ∎

References

[BEO20] Dave Benson, Pavel Etingof, and Victor Ostrik. New incompressible symmetric tensor categories in positive characteristic, 2020.

[CEO23] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. On Frobenius exact symmetric tensor categories. *Annals of Mathematics*, 197(3):1235–1279, 2023.

[CEO24] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. Incompressible tensor categories. *Advances in Mathematics*, 457:109935, 2024.

[CF24] Kevin Coulembier and Johannes Flake. Towards higher Frobenius functors for symmetric tensor categories, 2024. The course discussion follows the May 2024 version.

[Del02] Pierre Deligne. Catégories tensorielles. *Moscow Mathematical Journal*, 2(2):227–248, 2002.

[DM82] Pierre Deligne and James S. Milne. Tannakian categories. In *Hodge Cycles, Motives, and Shimura Varieties*, volume 900 of *Lecture Notes in Mathematics*, pages 101–228. Springer, 1982. Corrected version, 2022, with the original proposition numbering.

[EGNO15] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor categories*, volume 205 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2015.

[ENO05] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. On fusion categories. *Annals of Mathematics*, 162(2):581–642, 2005.

[EO19] Pavel Etingof and Victor Ostrik. On semisimplification of tensor categories, 2019.

[Mil12] James S. Milne. Basic theory of affine group schemes, 2012. Available at www.jmilne.org/math/.

[Ost15] Victor Ostrik. On symmetric fusion categories in positive characteristic, 2015.

[Wat79] William. C. Waterhouse. *Introduction to Affine Group Schemes*. Springer Graduate texts in mathematics. Springer New York, NY, 1979.

[Wib25] Michael Wibmer. A remark on torsors under affine group schemes. *Transformation Groups*, 30:447–454, 2025.